Autonomous Agricultural Drone Navigation System (Draft)

Prepared by: Team AeroAgri (Rohan Sharma, Maya Patel)

Abstract

This project proposes a drone platform for crop health monitoring. Edge algorithms on Jetson Nano detect leaf blight.

1. Problem Statement & Operational Challenge

Agricultural smallholders lose up to 32% of seasonal crop yields to late-detected fungal blights. Manual crop walking is labor-prohibitive across multi-acre terrains and cloud drones fail due to weak rural cellular connectivity.

System Architecture

- Aerial Platform: Carbon-fiber hexacopter with Pixhawk 4 autopilot running ArduPilot firmware.
- Edge Vision Pipeline: Raspberry Pi Global Shutter camera connected to Jetson Nano.

[TODO: Insert detailed Proposed Solution section, flight bench specifications, and Conclusion before submission]

- Aerial Platform: Carbon-fiber hexacopter with Pixhawk 4 autopilot running ArduPilot firmware.